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Effective removal of metal-carbide contamination from diamond circuit boards by reactive ion etching

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ABSTRACT

Recently, we have demonstrated the feasibility of using laser-patterned polycrystalline diamond sheets as a substrate for implantable circuit boards. The laser grooves in the diamond can be filled with biocompatible braze alloys such as Au-ABA and polished back to create long-lasting, hermetic circuit boards suitable for implantation in the human body. However, during the process of depositing molybdenum and niobium layers to aid adhesion of the Au-ABA braze, a residual film was found to form, which was resistant to both polishing and chemical etching, resulting in electrical shorting between features on the circuit board. Here, we report an optimised reactive ion etching process to remove the residual metal contamination and electrically isolate fine features on the diamond circuit boards. Samples were evaluated through energy dispersive spectroscopy elemental analysis and electrical testing of the diamond surface, confirming the complete removal of contaminants from the diamond surface.

1. Introduction

Diamond is emerging as a promising material for use in bioelectronics and medical implants owing to its longevity, chemical stability, and biocompatibility [1–11]. Moreover, diamonds possess a wide range of properties spanning from electrically conducting to highly insulating, from high surface area to ultra-smooth, and always with high thermal conductivity [12–19]. We have previously described different engineering approaches to creating diamond circuit boards [20–22], diamond hermetic packages [5,20,23] and various electrodes for neural stimulation [10,24,25], neural recording [11] and dopamine detection [11].

The diamond circuit board (DCB) under development in this work is a component in a brain implant for recording high-resolution neural activity from the cortex. The recording array is a pivotal component in an epilepsy management system currently under development by the medical device company Carbon Cybernetics [26]. Fig. 1 shows a magnified photograph of the circuit board in a 3D Printed transfer tray.

The array is produced by laser milling the diamond surface to form trenches and cavities, coating the surface with a braze alloy in paste form, melting the braze under controlled conditions, and polishing the surface to remove excess braze. Fig. 2 is an illustration of the process.

In previous work [5,21], we used silver active braze alloy (Ag-ABA, Ag 92.75 %, Cu 5 %, Al 1 %, Ti 1.25 %, Wesgo Ltd) to fill the trenches and form the interconnects. Ag-ABAs have excellent wetting properties on the diamond and form robust high conductivity circuit boards [21]. Silver, however, is toxic inside the body and unsuitable for use in implants [5]. An alternative is using gold active braze alloys (Au-ABAs), which exhibit excellent biocompatibility and biostability [4,5] but have poor wetting properties on the diamond surface, resulting in poor filling of milled features on the diamond substrates. To improve the wetting of Au-ABA on the diamond surface during the brazing process, we deposit a thin film (100-500 nm) of molybdenum (Mo) and niobium (Nb) [9]. The circuit board depicted in Fig. 1 was produced by depositing 200 nm of Mo followed by 200 nm of Nb before applying the braze paste.

Incorporating wetting layers of Mo and Nb enhanced the coverage of features and filling of features on the diamond circuit boards; we expected residual thin Mo and Nb to be removed by aggressive mechanical polishing (Fig. 2-c). However, the electrical shorting between two or more conductive tracks on the circuit board was often observed, leading

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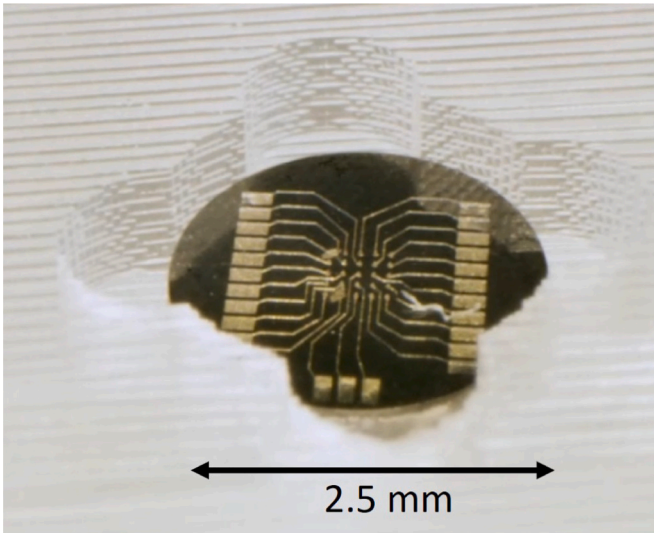

Fig. 1. Photograph of a diamond circuit board substrate for a high-resolution cortical recording electrode array.

to malfunction or failure of the device [27–29]. The observation of shorting between the interconnects indicated that a residual layer remained, probably in the form of metal carbides. This layer proved to be highly resistant to removal either by chemical and plasma etching or mechanical polishing. Here we report optimised recipes for a reactive ion etching (RIE) technique to selectively remove these contamination layers rapidly and efficiently while not damaging the Au braze. The process eliminates the residual metal-carbide contamination and electrically isolates features on the diamond circuit boards resulting in functional diamond circuit boards.

RIE has been proposed as an alternative method for cleaning diamond surfaces due to its ability to remove various contaminants, including metal particles [30–39]. RIE is a dry etching process that uses reactive gases to remove material chemically and physically from the surface of a substrate [35,40–42]. The process is highly selective, meaning it can remove specific materials while leaving others intact without damaging the material's surface, leading to a clean surface that enhances the electrical properties of the material [43–45]. A range of gas mixtures and etch times were tested. The samples were evaluated using a combination of analytical techniques to characterise the surface before and after cleaning. Our study demonstrates that the optimised reactive ion etching process successfully removes metal contamination, offering a solution to the persistent contamination issue encountered when using metallic wetting layers on diamond surfaces.

2. Materials and method

2.1. Sample preparation (diamond circuit board fabrication and Au-ABA brazing)

The process for preparing diamond circuit boards with Mo/Nb has been previously described [9]. Briefly, TM100 polycrystalline CVD diamond (PCD) plates sourced from Element Six Ltd. [46] with a thickness of 0.3 mm and single-sided polishing (<50 nm RMS) were used as the substrate material. Gold active braze alloy (Au-ABA), which contained 96.4 % Au, 3 % Ni, and 0.6 % Ti as per the manufacturer's specifications [47], was purchased from Morgan Advanced Materials [Hayward, CA, USA] in paste form. Diamond samples were patterned using a 2.5 W Nd:YAG, 532 nm wavelength, nanosecond pulsed laser micromachining system (Oxford Laser). We used two distinct patterns for the samples: Sample A, which is illustrated in Fig. 1, is a circuit board designed to be used in a neural recording device, and Sample B, which used larger tracks and was used mainly for ease of analysis. The samples were cleaned by boiling them for 60 min in an acid solution of $NaNO_2$/H_2SO_4. A bilayer of 200 nm of Mo followed by 200 nm Nb was deposited on the diamond substrate via DC sputtering using an Intlvac Nanochrome AC/DC Sputter System, with a vacuum pressure of 2×10^{-3} Torr. Au-ABA paste was manually added to the substrates and binder dried on a hotplate at 110 °C for 10–15 min. Subsequently, the samples were brazed in an MTI brazing furnace at 1100 °C for 30 min under a vacuum of better than 10^{-5} Torr with a ramp-up time of 360 min and a ramp-down time of 600 min. Finally, the excess brazes removed by polishing the samples using a Coborn PL3 rotary polisher. A 600-grit polishing wheel was used for between 1 and 3 min.

2.2. Reactive ion etching process for removal of contamination

The employed RIE plasma system (DRIE Multi Oxford Plasma Lab100) is a commercial parallel plate reactor. To remove the metallic contamination from the diamond circuit boards, we utilised RIE with various gas compositions of Ar/Cl_2, CF_4/O_2, and $CF_4/Cl_2/Ar$ and different etching times. Details regarding the three different gas mixtures and RIE conditions are provided in Table 1. These gas compositions were reported as common effective RIE reactants for removing single/polycrystalline diamond [30,31,34,35,48,49] and metal (Mo, Nb, Ti) contaminations [38,50–54]. So, we selected mentioned compositions to

Table 1
Gas mixture and RIE conditions.

Recipe #	Gas composition	Gas flow (sccm)	Bias/ICP Power (W)	Process pressure (mTorr)	Etching time (min)
1	Ar/Cl_2	5/30	100/1000	5	20–80
2	CF_4/O_2	5/30	100/1000	5	5–30
3	$CF_4/Cl_2/Ar$	5/10/40	100/1000	15	20–80

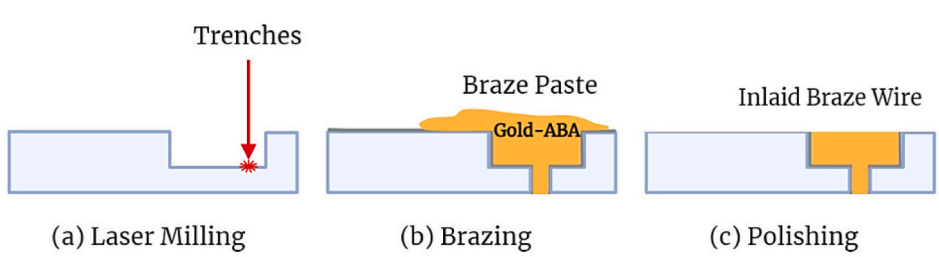

Fig. 2. Illustration of forming inlaid braze wire circuit boards in diamond. Created with BioRender.com, adapted from [9].

investigate their effects on the low pressure and the same Bias (RF) and ICP powers, varying the etching time between 5 and 80 min.

2.3. Sample characterisation

Samples were characterised by scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDX) (SEM/EDX Hitachi TM), and electrical measurements before and after cleaning.

2.4. Electrical characterisation of samples

Electrical characterisation was measured before and after the RIE process to demonstrate electrical shorting across the diamond surface and between tracks on the circuit board. Fig. 3-a shows a microscope photograph of a diamond/Au-ABA circuit board, in a holder, with tungsten microprobes in contact with the surface. Conductivity measurements were recorded between a ground pad indicated in Fig. 3 and multiple locations on the surface of the diamond (points 1 to 3 in Fig. 3-b) and between adjacent brazed pads (point 4 in Fig. 3-b).

3. Results and discussion

3.1. SEM and EDX analysis

Samples were analysed before and after RIE treatment to understand the direct effects. Fig. 4-a and b shows the EDX analysis of a sample just after polishing and before RIE. Fig. 4-b shows a high concentration of undesired metals (Mo and Ti) between the tracks and pads. Mikhailov et al. have previously reported that molybdenum carbide is formed between the diamond substrate and sputtered Mo films at temperatures above 550 °C [55]. Hence, it is likely that polishing-resistant Mo is a carbide as well as titanium contamination. The presence of Ti is surprising as its only possible source was from the braze, and 400 nm of Mo and Nb lie between the braze filler and the diamond substrate. However, TiC is known to form at temperatures as low as 25 °C [9] and we presume the small amount of titanium in the Au-ABA can diffuse through the Mo/Nb wetting layers and interact with the diamond surface [9].

Our previous article [9] described that Nb is more soluble than Mo in Au and Ni. This is a possible reason why Nb has diffused into the Au-ABA more than Mo. Furthermore, although Ti has higher solubility in Au and Ni than Mo and Nb, it migrated to the diamond interface to form TiC. This is likely because the Gibbs standard free energy of the formation of TiC is much lower than the other carbides of Mo or Nb [9].

TiC and Mo carbides, such as Mo_2C, are hard ceramic materials that are difficult to polish due to their high hardness and chemical inertness [56–59]. Traditional polishing methods may not be effective, and chemical polishing can also be challenging. TiC and Mo_2C are chemically stable, not reacting readily with most acids, bases, and organic solvents [60,61]. Specialised techniques have been developed for polishing hard materials, such as ion beam milling, magnetorheological finishing, and plasma-assisted polishing.

Mo and Ti contamination led to the electrical shorting of the gold-brazed pads. Thus, it is essential to remove these materials effectively. Multiple RIE recipes were tested to clean the samples. Of the RIE recipes tested, the O_2/CF_4 recipe (Recipe 2 in Table 1) was the most effective in removing contamination in the least amount of time. Fig. 4-c and d shows SEM images and EDX maps of a diamond circuit board after RIE treatment. A clean diamond surface and intact gold-brazed features are obtained after only 30 min of treatment.

Recipes 1 and 3 were also practical, but longer etching times were required to eradicate all contamination. Upon examining Recipe 1 (Ar/Cl_2) as an etchant for gold [53,54], it has been established that while it is effective, it also presents a challenge in that the bombardment by Ar ions removed some of the gold brazed layers, particularly in shallow areas, as shown in Fig. S1. Using Recipe 2 (CF_4/O_2), shorter times (5 and 10 min) were tested on a separate, highly contaminated sample. Fig. 5 shows EDX results around the brazed pads before etching (Fig. 5-a), after 5 min of etching (Fig. 5-b) and after 10 min of etching (Fig. 5-c).

In Fig. 5-b, contamination is still visible on the EDX images after 5 min of etching. However, the samples appear clean after 10 min. A qualitative assessment of the three RIE recipes tested is shown in Table 2.

3.2. Mechanism of metal contamination removal by RIE

This mechanism of removal by RIE is due to the ability of the reactive ions in the plasma to chemically interact with and selectively etch/remove/sputter the metal particles or films [34,35,40,41]. The carbide films/particles are broken down into smaller molecules, physically sputtered out by the reactive ions [45]. The chemical reactivity of the metal particles with the reactive ions in the plasma allows them to be selectively removed from the surface of the diamond substrate [44,45]. The specific conditions of the RIE process, such as the gas mixture, pressure, and power settings, can be tailored to optimise the removal of metal contamination from diamond surfaces [35,45,62].

Applying electrical power to a low-pressure gas generates plasma by

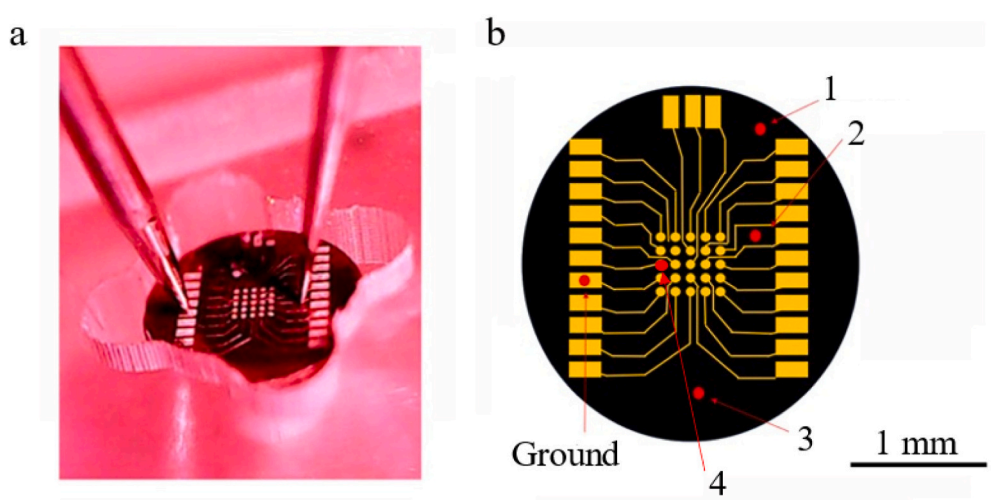

Fig. 3. a) Photograph of diamond/gold ABA circuit board with microprobes in place. b) Illustration of probe placement locations.

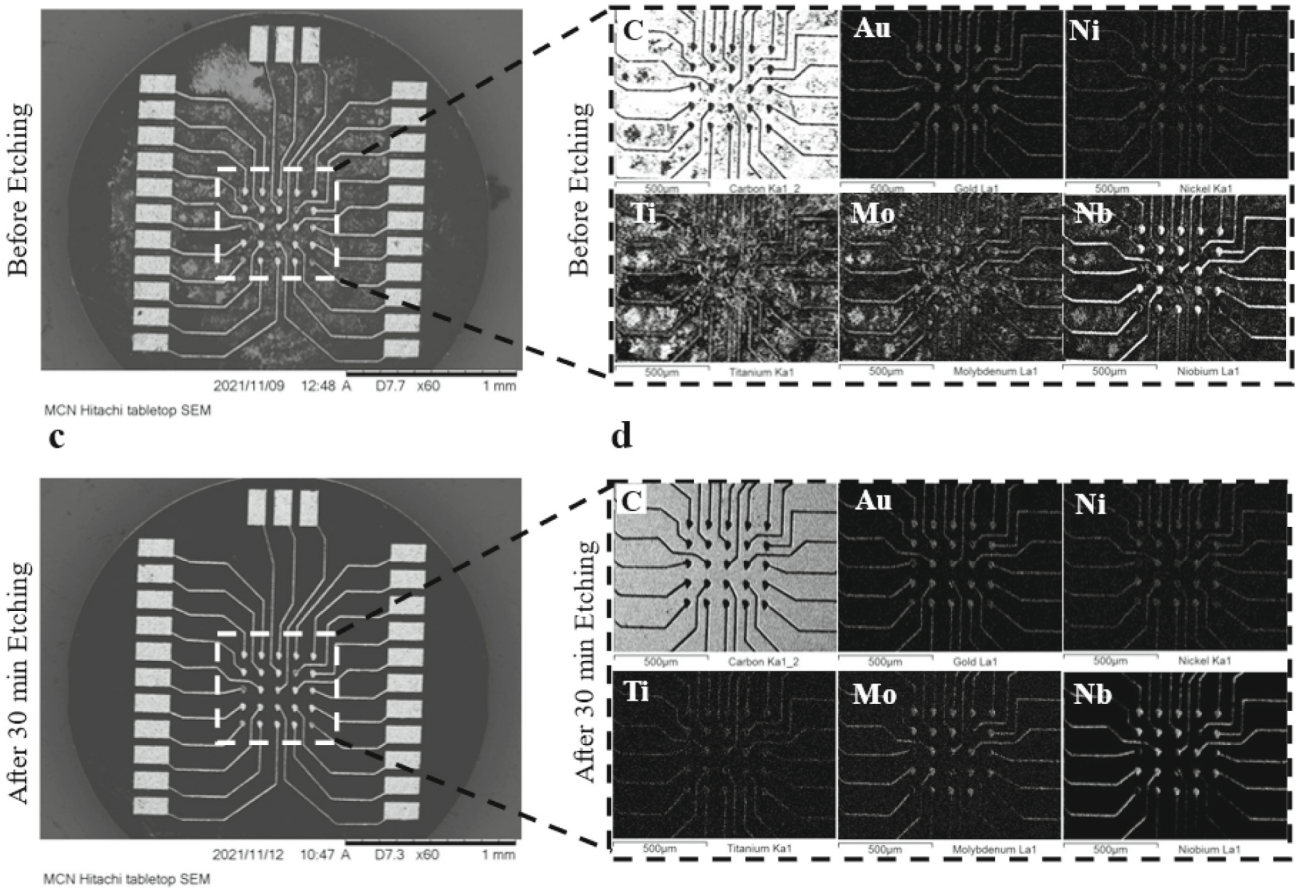

Fig. 4. a) SEM image of sample A before the RIE process, b) EDX analysis of elements of C, Au, Ni, Ti, Mo and Nb of the middle part of sample A (dashed square in (a)), c) SEM image of sample A after 30 min of etching with O_2/CF_4, and d) EDX analysis of elements of C, Au, Ni, Ti, Mo and Nb of the middle part of sample A (dashed square in (c)).

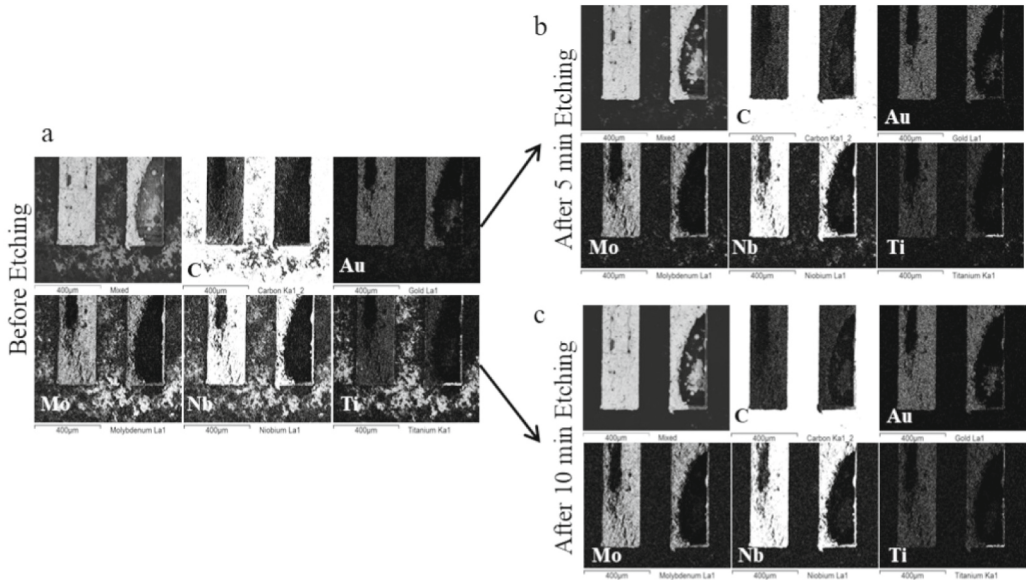

Fig. 5. EDX analysis of sample B a) before the RIE process, b) after 5 min of etching with O_2/CF_4, and c) after 10 min of etching with O_2/CF_4.

transferring energy to electrons, leading to complex chemical reactions in the plasma phase [63]. These reactions can cause material etching or the formation of a thin film on the sample surface, depending on the chemical affinity and volatility of the product [63]. As mentioned earlier, here we considered Mo and Ti contamination as carbides as well as metal particles to investigate possible reactions inside of the plasma during RIE treatment. The dry etching process operates through a reaction mechanism where the etchant gas molecules are first dissociated

Table 2
A qualitative assessment of the three RIE recipes tested.

Recipe	Composition	Rank	Observations	Etch time
1	Ar/Cl_2	Worst	Gold etching & longer time	20–80
2	CF_4/O_2	Best	Optimised etching time	5–30
3	$CF_4/Cl_2/Ar$	Average	No gold etching but a longer time	20–80

in the plasma and then undergo a chemical reaction with the sample's surface [45]. In the CF_4/O_2 gas mixture, CF_4 molecules dissociate in the plasma, producing F and CF_3 radicals, Eq. (1) (CF_4* represents CF_4 in an excited state).

$$CF_4 \rightarrow CF_4^{*} \rightarrow CF_3^{\cdot} + F \tag{1}$$

Oxygen gas (O_2) is utilised as a co-reactant, which aids in eliminating the reaction byproducts from the surface, facilitating an uninterrupted etching process. These CF_3 and F radicals react with the metal layer on the diamond circuit board, such as TiC, Mo carbides or Nb, and form volatile fluorides (Eqs. (2) to (11)) and/or COF_2. The reaction products then desorb from the surface and are pumped out of the chamber. Possible reactions are as follows:

$$TiC + 2O_2 + 2CF_4 \rightarrow TiO_2 + 2COF_2 \tag{2}$$

$$TiC + 4CF_4 + 2O_2 \rightarrow TiF_4 + 4COF_2 + CO_2 \tag{3}$$

$$Mo_2C + 3O_2 + 4CF_4 \rightarrow 2MoO_3 + 4COF_2 \tag{4}$$

$$Mo_2C + 6CF_4 + 3O_2 \rightarrow 2MoF_6 + 6COF_2 + CO_2 \tag{5}$$

$$Nb + 5CF_4 + 5O_2 \rightarrow NbF_5 + 10COF_2 \tag{6}$$

$$Nb + 2CF_4 + 2O_2 \rightarrow NbO_2 + 4COF_2 \tag{7}$$

$$Ti + 4F^{\cdot} \rightarrow TiF_4 \tag{8}$$

$$Ti + xF^{\cdot} \rightarrow TiF_x + CF_4 \tag{9}$$

$$Mo + 4F^{\cdot} \rightarrow MoF_4 \ (MoF_4 + O_2 \rightarrow MoO_3) \tag{10}$$

$$Nb + 5F^{\cdot} \rightarrow NbF_5 \ (NbF_5 + O_2 \rightarrow Nb_2O_5) \tag{11}$$

One possible reaction between TiC and CF_4/O_2 gas mixture is the formation of a TiO_2 layer on the surface of TiC, which can act as a passivation layer, slowing down the etching rate (Eq. (2)). Another possible reaction is the formation of TiF_4 and CO_2 (Eq. (3)). Possible reactions between Mo_2C and CF_4/O_2 gas mixture are the formation of a MoO_3 layer on the surface of Mo_2C (Eq. (4)) or the formation of MoF_6 and CO_2 (Eq. (5)). In these reactions (Eqs. (2) to (11)), TiF_x, MoF_x, and NbF_5 are volatile compounds (Table 3) that are removed from the surface, while CF_4 is also volatile and contributes to the overall etching process [63].

In the case of using Cl_2 and Ar for etching diamonds, Ar ions are used to sputter the diamond surface and facilitate the reaction between Cl_2 and carbon, resulting in the formation of volatile CCl_x products [35]. Ar serves as the primary active etching species, functioning similarly to an ion milling process where ions collide with the surface and break bonds through physical means [31]. It has been reported that the transition from a fluorine-based to a chlorine-based etching procedure leads to lower etch rates and a need for increased ion assistance, resulting in a more easily controlled etching profile due to reduced volatility of etch byproducts [63]. Table 3 provides a list of various halogen-compounds, and their boiling temperatures for the most common elements and materials that plasma etching deals with shortly.

Table 3
Halogen -compounds and their boiling temperatures for elements and materials.

Elements	Fluorides	Boiling temperature (°C)	Chlorides	Boiling temperature (°C)
C	CF_4	−128	CCl_4	77
Ti	TiF_4	284 (subl.)	$TiCl_4$	136.4
Mo	MoF_5	213.6	$MoCl_5$	268
	MoF_6	35	$MoOCl_3$	100 (subl.)
	MoO_2F_2	270 (subl.)		
	$MoOF_4$	180		
Nb	NbF_5	236	$NbCl_5$	248.2

It is noteworthy to mention that the semi-empirical correlation between the boiling temperature and vapor pressure of a substance is described widely by the Antoine equation [64,65]. It is expressed as Eq. (12)

$$Log\ (P) = A - \left(\frac{B}{T + C}\right) \tag{12}$$

where P is the vapor pressure in mmHg, T is the temperature in degrees Celsius, and A, B, and C are substance-specific constants [64,65]. The Antoine equation shows that as temperature increases, vapor pressure increases exponentially due to more molecules escaping into the gas phase. The equation helps predict vapor pressure at different temperatures and has practical applications in chemical processes, thermodynamics, and engineering design [64–66]. Therefore, at a given temperature, substances with a higher vapor pressure are more volatile and have lower normal boiling temperature. Consequently, substances with lower boiling temperature demonstrate greater volatility. According to the information presented in Table 3, fluorides generally have more significant vapor pressures than chlorides. This higher vapor pressure suggests that the etching byproducts of fluorides are more volatile than those of chlorides. Therefore, when using a gas chemistry that produces fluorides, the resulting etching byproducts can more easily desorb, resulting in a higher etch rate [45]. Accordingly, using CF_4/O_2 as reactant results in a higher etch rate compared to using Ar/Cl_2 etchant.

3.3. Electrical characterisation

Electrical characterisation was conducted to demonstrate that electrical shorting across the diamond surface and between tracks on the circuit board was reduced by RIE cleaning. Table 4 shows the resistivity measurements recorded across points 1 to 4, and the ground pad is shown in Fig. 3-b before and after RIE.

Before RIE, the resistance across the diamond surface was measured at approximately 50 kΩ from all 3 locations; this value indicates electrical shorting across the surface of the diamond [67]. Following RIE, these locations' resistance increased drastically to 8.75×10^{11} Ω. The resistance between adjacent wires on the circuit board was measured at 3.37×10^{11} Ω. Isolation in the Giga ohm range is standard for data transmission in the 10–25 kHz range. For instance, Omnetics connectors, commonly used in electrophysiology, specify line isolation higher than 5 GΩ for their commercial grade connectors [67].

4. Conclusion

RIE using three gas mixtures was found to remove metal and metal

Table 4
Resistance results before and after RIE using recipe 2 for points 1 to 4 are shown in Fig. 3-b.

Probe location	Before RIE (Ω)	After RIE (Ω)
1	5.12×10^4	3.31×10^{11}
2	5.15×10^4	7.75×10^{11}
3	5.13×10^4	8.75×10^{11}
4		3.37×10^{10}

carbide contamination following the brazing of diamond with Au based braze alloys. This contamination was found to shorten the contacts, and its removal was essential for the proper functioning of the diamond circuit boards. SEM and EDX were used to assess the degree to which contamination was removed. Although different RIE gas mixtures were effective in reducing contamination, the best results were obtained using a CF_4/O_2 gas mixture. Electrical testing confirmed that electrical shorting in the 50 kΩ range was eliminated by CF_4/O_2 RIE cleaning resulting in isolation between tracks in the GΩ range, which is standard for data transmission.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.diamond.2023.110329.

CRediT authorship contribution statement

Khatereh Edalati, RIE experiments, Sample preparation, EDS, SEM, Manuscript preparation. *Simon Higham*, Electrical characterisation, Sample preparation. *Sorel E. De Leon*, Characterisation SEM training. *Shun Long Cyril Au*, Sample preparation, Methodology. *Athavan Nadarajah*, RIE training and optimisation. *Steven Prawer*, Conceptualisation Methodology, Editing. *David J. Garrett*, Conceptualisation, Methodology, Editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: David J Garrett reports financial support was provided by ARC. David J Garrett, reports a relationship with Carbon Cybernetics Pty Ltd. that includes: equity or stocks. Steven Prawer reports a relationship with ARC that includes: board membership and equity or stocks. Shun Long Cyril Au reports a relationship with Carbon Cybernetics Pty Ltd. that includes: employment and equity or stocks.

Data availability

Data will be made available on request.

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